Many Daughters Pre-Analysis Plan

Hypothesis 1

H1. *Having daughters leads to stronger support for gender equality (Y_1) (as measured by attitudes).*

(Y_1 for respondents with daughters) > (Y_1 for respondents without daughters).

Variables used to test H1:

Variable name	**Function**	**SOEP variable**	**SOEP data set**	**SOEP Info**	**Transformation**
y_{1i} (Support for gender equality)	Outcome	plh0308_v2	pl	"Best if man and woman work the same amount so they can share the responsibility"	None: Measured on 7-point Likert scale: 1 (Completely Disagree) to 7 (Fully Agree)
d_i (Having biological daughter(s))	Treatment	kidsex*	biobirth	Sex of *first, second, …. * child	Transformed: Equals 1 if parent has only biological daughters and 0 if only biological sons
$\boldsymbol{X_{1i}}$ (Birthyear of first child fixed effects)	Control variable	kidgeb01	biobirth	Year of Birth 1st Child	Transformed: Vector of dummy variables equal to 1 if first child born in specific year, 0 otherwise
$\boldsymbol{X_{2i}}$ (Birthyear of parent i fixed effects)	Control variable	gebjahr	biobirth	Birth Year	Transformed: Vector of dummy variables equal to 1 if parent born in specific year, 0 otherwise
x_{3i} (Migration background of parent i)	Control variable	migback	ppathl	Migration background	Transformed: Equals 1 if parent has direct or indirect migration background, 0 if not
First child ID	Cluster variable	kidpnr01	biobirth	Person identifier 1st child	None
Individual weights	Weighting variable	phrf	ppathl	Personal weighting factor	None

Sample used to test H1:

Data source: SOEP-Core

Unit of observation: Parent i

Selection Criteria:

- All parents whose biological children are all of the same sex based on kidsex* from the biobirth data set
- Restrict to survey year 2018 (as the outcome is only available this year) based on syear from the ppathl data set
- Restrict to parents whose first child was born before the survey year and whose first child is below 18 years old based on kidgeb01 (biobirth) and syear (ppathl)
- Complete case analysis: Only consider parents which have no missing for any of the relevant variables: plh0308_v2, kidsex*, kidgeb01, gebjahr, migback, kidpnr01, phrf

Analysis used to test H1:

To test H1, we will estimate the following linear regression model:

$$y_{1i} = \alpha + \beta d_i + X_{1i}\boldsymbol{\gamma_1} + X_{2i}\boldsymbol{\gamma_2} + x_{3i}\gamma_3 + \varepsilon_i \qquad (1)$$

y_{1i}: Support for gender equality (outcome variable)

d_i: Dummy indicating whether parent i has only biological daughters

X_{1i}: Vector of fixed effects for the birth year of the first child

X_{2i}: Vector of fixed effects for the parent's birth year

x_{3i}: Indicator for whether the parent has a migration background

ε_i: Error term

Clustered standard errors: Standard errors will be clustered at the level of the first child using the identifier kidpnr01 from the biobirth data set

Sampling weights: We will run a weighted least squares regression using the weighting variable phrf from the ppathl data set, which reflects the SOEP's individual-level survey weights

Decision criteria used for H1:

We will test the hypothesis using the coefficient β in regression equation (1).

Statistical significance will be assessed using a one-sided t-test at the 5% significance level. A positive and statistically significant estimate of β will be interpreted as evidence that having daughters leads to stronger support for gender equality.

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Hypothesis 2

H2. *Having daughters leads to more left-leaning political preferences (Y_2).*

(Y_2 for respondents with daughters) > (Y_2 for respondents without daughters

Variables used to test H2:

Variable name	**Function**	**SOEP variable**	**SOEP data set**	**SOEP Info**	**Transformation**
y_{2it} (Political preferences)	Outcome	plh0004	pl	"… If you think about your own political views: Where would you place yours?"	Transformed: The original ordinal scale (0 far left, …, 10 far right) will be reversed by subtracting each value from 10
d_i (Having biological daughter(s))	Treatment	kidsex*	biobirth	Sex of *first, second, …. * child	Transformed: Equals 1 if parent has only biological daughters and 0 if only biological sons
$\boldsymbol{X}_{1i}$ (Birthyear of first child fixed effects)	Control variable	kidgeb01	biobirth	Year of Birth 1st Child	Transformed: Vector of dummy variables equal to 1 if first child born in specific year, 0 otherwise
$\boldsymbol{X_{2i}}$ (Birthyear of parent i fixed effects)	Control variable	gebjahr	biobirth	Birth Year	Transformed: Vector of dummy variables equal to 1 if parent born in specific year, 0 otherwise
x_{3i} (Migration background of parent i)	Control variable	migback	ppathl	Migration background	Transformed: Equals 1 if parent has direct or indirect migration background, 0 if not
$\boldsymbol{X_{4t}}$ (Survey year fixed effects)	Control variable	syear	ppathl	Survey Year	Transformed: Vector of dummy variables equal to 1 if observation is from survey year t, 0 otherwise
First child ID	Cluster variable	kidpnr01	biobirth	Person identifier 1st child	None
Individual weights	Weighting variable	phrf	ppathl	Personal weighting factor	None

Sample used to test H2:

Data source: SOEP-Core

Unit of observation: Parent i in survey year t

Selection Criteria:

- All parents whose biological children are all of the same sex based on kidsex* from the biobirth data set
- Restrict to survey years 2005, 2009, 2014, 2019 (as the outcome is only available these years) based on syear from the ppathl data set
- Restrict to parent-year observations where the first child was born before the respective survey year based on kidgeb01 (biobirth) and syear (ppathl)
- Restrict to parent-year observations where the first child is under 18 years old in the respective survey year based on kidgeb01 (biobirth) and syear (ppathl)
- Complete case analysis: Only consider observations which have no missing for any of the relevant variables: plh0004, kidsex*, kidgeb01, gebjahr, migback, syear, kidpnr01, phrf

Analysis used to test H2:

To test H2, we will estimate the following linear regression model:

$$y_{2it} = \alpha + \beta d_i + X_{1i}\boldsymbol{\gamma_1} + X_{2i}\boldsymbol{\gamma_2} + x_{3i}\gamma_3 + X_{4t}\boldsymbol{\gamma_4} + \varepsilon_{it} \qquad (2)$$

y_{2it} : Left leaning political preference

d_i: Dummy indicating whether parent i has only biological daughters

X_{1i}: Vector of fixed effects for the birth year of the first child

X_{2i}: Vector of fixed effects for the parent's birth year

x_{3i}: Indicator for whether the parent has a migration background

X_{4t}: Vector of survey year fixed effects

ε_{it}: Error term

Clustered standard errors: Standard errors will be clustered at the level of the first child using the identifier kidpnr01 from the biobirth data set

Sampling weights: We will run a weighted least squares regression using the weighting variable phrf from the ppathl data set, which reflects the SOEP's individual-level survey weights

Decision criteria used for H2:

We will test the hypothesis using the coefficient β in regression equation (2).

Statistical significance will be assessed using a one-sided t-test at the 5% significance level. A positive and statistically significant estimate of β will be interpreted as evidence that having daughters leads to more left-leaning political preferences.

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Hypothesis 3

H3. Having daughters leads to more gender equality in the household (Y_3) (as measured by behaviors or outcomes).

(Y_3 for respondents with daughters) > (Y_3 for respondents without daughters)

Variables used to test H3:

Variable name	**Function**	**SOEP variable**	**SOEP data set**	**SOEP Info**	**Transformation**
y_{3ct} (Gender equality in the household)	Outcome	- Housework: pli0043_h, pli0012_h, pli0016_h - Childcare: pli0044_h, pli0019_h, pli0022_h - Repairs: pli0049_h, pli0031_h, pli0034_v4	pl	*"… How many hours do you spend on the following activities on a typical weekday, Saturday, and Sunday?"*	Weekly hours spent on housework, childcare, and repairs are summed for each parent (weekday hours * 5, plus Saturday, plus Sunday). Then, the following gender equality index is calculated: $y_{3ct} = 1 - \left\lvert \frac{\text{Hours}_{\text{woman}}}{\text{Hours}_{\text{woman}} + \text{Hours}_{\text{man}}} - 0.5 \right\rvert$ The index ranges from 0 (maximum inequality) to 1 (perfect equality) in time allocation between partners.
d_c (Having biological daughter(s))	Treatment	kidsex*	biobirth	Sex of *first, second, …. * child	Transformed: Equals 1 if couple has only biological daughters and 0 if only biological sons
$\boldsymbol{X_{1c}}$ (Birthyear of first child fixed effects)	Control variable	kidgeb01	biobirth	Year of Birth 1st Child	Transformed: Vector of dummy variables equal to 1 if first child born in specific year, 0 otherwise
$\boldsymbol{X_{4t}}$ (Survey year fixed effects)	Control variable	syear	ppathl	Survey Year	Transformed: Vector of dummy variables equal to 1 if observation is from survey year t, 0 otherwise
First child ID	Cluster variable	kidpnr01	biobirth	Person identifier 1st child	None
Couple weights	Weighting variable	phrf	ppathl	Personal weighting factor	Transformed: Take the average of the man's and the woman's individual weighting factor

Sample used to test H3:

Data source: SOEP-Core

Unit of observation: Couple c in survey year t

Selection Criteria:

- Consider heterosexual couples who have their first child together and live together in the same household in survey year t based on kidpnr01 (biobirth) and hid (ppathl)

- Exclude couples in which at least one parent has biological children of a different sex than their firstborn based on kidsex* from the data set biobirth

- Restrict to survey years 1993, 1995, 1997, 1999, 2001, 2003, 2005, 2007, 2009, 2011, 2013, 2015, 2017, 2019, 2021 (as the outcome is only available these years) based on syear from the ppathl data set

- Restrict to couple-year observations where the first child was born before the respective survey year based on kidgeb01 (biobirth) and syear (ppathl)

- Restrict to couple-year observations where the first child is under 18 years old in the respective survey year kidgeb01 (biobirth) and syear (ppathl)

- Complete case analysis: Only consider couples for which both partners do not have missing values for any of the relevant variables: pli0043_h, pli0012_h, pli0016_h, pli0044_h, pli0019_h, pli0022_h, pli0049_h, pli0031_h, pli0034_v4, kidsex*, kidgeb01, syear, kidpnr01, hhrf

Analysis used to test H3:

To test H3, we will estimate the following linear regression model:

$$y_{3ct} = \alpha + \beta d_c + X_{1c}\boldsymbol{\gamma_1} + X_{4t}\boldsymbol{\gamma_4} + \varepsilon_{ct} \qquad (3)$$

y_{3ct}: Gender equality in the household

d_c: Dummy indicating whether couple c has only biological daughters

X_{1c}: Vector of fixed effects for the birth year of the first child

X_{4t}: Vector of survey year fixed effects

ε_{ct}: Error term

Clustered standard errors: Standard errors will be clustered at the level of the first child using the identifier kidpnr01 from the biobirth data set

Sampling weights: We will run a weighted least squares regression using the mean of the weighting variable phrf for the man and the woman of the couple from the ppathl data set as the analysis is at the couple level and we need a weighting variable at the couple level.

Decision criteria used for H3:

We will test the hypothesis using the coefficient β in regression equation (3).

Statistical significance will be assessed using a one-sided t-test at the 5% significance level. A positive and statistically significant estimate of β will be interpreted as evidence that having daughters leads to more equality in the household.

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Hypothesis 4

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*

(Y_1 for male respondents with daughters - Y_1 for male respondents without daughters)

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(Y_1 for female respondents with daughters - Y_1 for female respondents without daughters)

Variables used to test H4:

Variable name	**Function**	**SOEP variable**	**SOEP data set**	**SOEP Info**	**Transformation**
y_{1i} (Support for gender equality)	Outcome	plh0308_v2	pl	"Best if man and woman work the same amount so they can share the responsibility"	None: Measured on 7-point Likert scale: 1 (Completely Disagree) to 7 (Fully Agree)
d_i (Having biological daughter(s))	Treatment	kidsex*	biobirth	Sex of *first, second, …. * child	Transformed: Equals 1 if parent has only biological daughters and 0 if only biological sons
m_i (Sex of parent is male)	Moderator	sex	biobirth	Gender	Transformed: Equals one if the parent is male and zero if female
$d_i \times m_i$ (Male parent with daughters)	Interaction Term	-	-	-	Transformed: Interaction between the two transformed variables d_i and m_i
$\boldsymbol{X_{1i}}$ (Birthyear of first child fixed effects)	Control variable	kidgeb01	biobirth	Year of Birth 1st Child	Transformed: Vector of dummy variables equal to 1 if first child born in specific year, 0 otherwise
$\boldsymbol{X_{2i}}$ (Birthyear of parent i fixed effects)	Control variable	gebjahr	biobirth	Birth Year	Transformed: Vector of dummy variables equal to 1 if parent born in specific year, 0 otherwise
x_{3i} (Migration background of parent i)	Control variable	migback	ppathl	Migration background	Transformed: Equals 1 if parent has direct or indirect migration background, 0 if not
First child ID	Cluster variable	kidpnr01	biobirth	Person identifier 1st child	None
Individual weights	Weighting variable	phrf	ppathl	Personal weighting factor	None

Sample used to test H4:

Data source: SOEP-Core

Unit of observation: Parent i

Selection Criteria:

- All parents whose biological children are all of the same sex based on kidsex* from the biobirth data set
- Restrict to survey year 2018 (as the outcome is only available this year) based on syear from the ppathl data set
- Restrict to parents whose first child was born before the survey year and whose first child is below 18 years old based on kidgeb01 (biobirth) and syear (ppathl)
- Complete case analysis: Only consider parents which have no missing for any of the relevant variables: plh0308_v2, kidsex*, kidgeb01, gebjahr, migback, kidpnr01, phrf

Analysis used to test H4:

To test H4, we will estimate the following linear regression model:

$$y_{1i} = \alpha + \beta_1 d_i + \beta_2 m_i + \delta d_i \times m_i \; + X_{1i}\boldsymbol{\gamma_1} + X_{2i}\boldsymbol{\gamma_2} + x_{3i}\gamma_3 + \varepsilon_i \qquad (4)$$

y_{1i} : Support for gender equality (outcome variable)

d_i: Dummy indicating whether parent i has only biological daughters

m_i: Dummy indicating whether parent i is male

$d_i \times m_i$: Interaction term equal to 1 if parent i is male and has only biological daughters; 0 otherwise

X_{1i}: Vector of fixed effects for the birth year of the first child

X_{2i}: Vector of fixed effects for the parent's birth year

x_{3i}: Indicator for whether the parent has a migration background

ε_i: Error term

Clustered standard errors: Standard errors will be clustered at the level of the first child using the identifier kidpnr01 from the biobirth data set

Sampling weights: We will run a weighted least squares regression using the weighting variable phrf from the ppathl data set, which reflects the SOEP's individual-level survey weights

Decision criteria used for H4:

We will test the hypothesis using the coefficient δ in regression equation (4).

Statistical significance will be assessed using a one-sided t-test at the 5% significance level. A positive and statistically significant estimate of δ will be interpreted as evidence that the effect of having daughters on support for gender equality is stronger for men than for women.